S3.1 The periodic table: Classification of elements

Structure of the periodic table

Learning outcomes

By the end of this section you should be able to:

- Identify the positions of metals, metalloids and nonmetals in the periodic table.
- Identify the block an element belongs to.
- Identify the group and period number of an element.

If you look at the two data tables in **Figure 1** which table do you immediately notice as being more useful? Why is this?

Data table 1

130	180.5
160	97.79
200	63.38
215	39.30

Data table 2

Element	Atomic radius ($\times10^{-12}$ m)	Melting point (°C)
Li	130	180.5
Na	160	97.79
K	200	63.38
Rb	215	39.30

Figure 1. Comparison of two data tables.

In general, tables are more useful if row and column headings are included. This is certainly true for the periodic table. While the periodic table has more information within each of its entry locations, it is incredibly useful to use row and column headings too.

Groups and periods

As Dmitri Mendeleev constructed his version of the periodic table, he arranged certain elements together based on similar properties. These elements now make up the columns of the periodic table we know today and each column is referred to as a group. There are 18 groups in the periodic table, starting with group 1 (H, Li, Na, K, Rb, Cs, Fr) and ending with group 18 (He, Ne, Ar, Kr, Xe, Rn, Og).

The modern periodic table is arranged horizontally by atomic number. Each row is referred to as a period, as each subsequent row shows similar properties of the elements, hence the term periodic. **Figure 2** shows how groups and periods are deduced in the periodic table. Additionally, the periodic table (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) in your DP chemistry data booklet has the group and period numbers labelled.

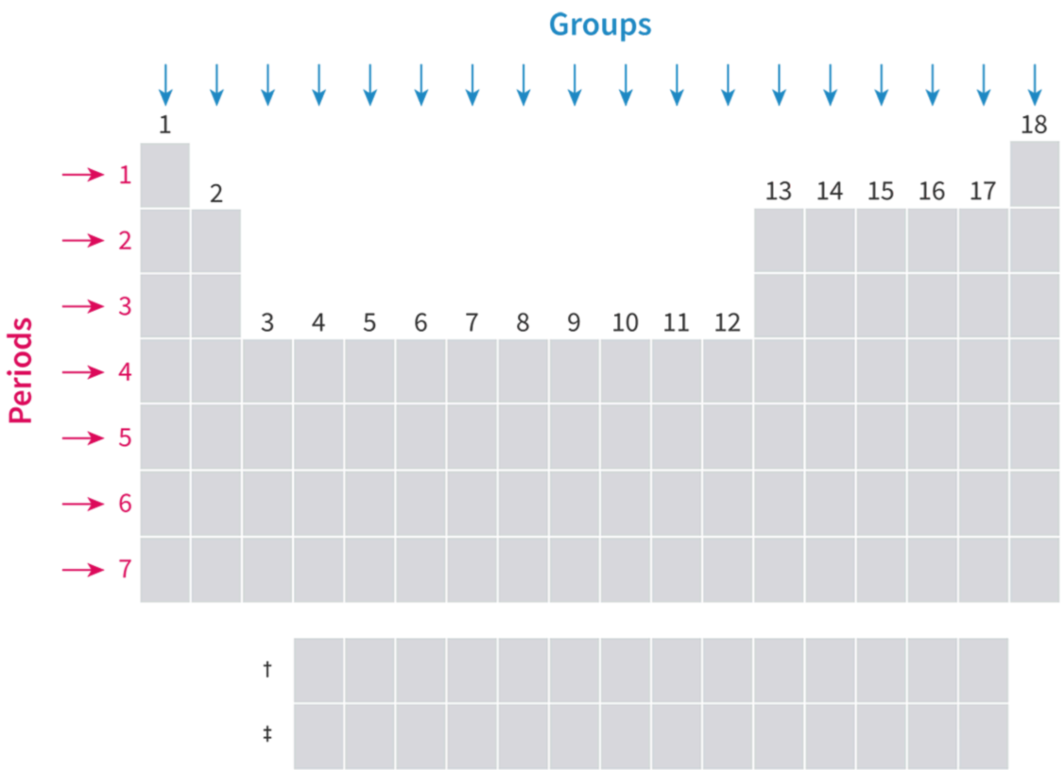

Figure 2. Group and period labels in the periodic table.

More information for figure 2

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The image is a section of the periodic table that specifically highlights the labels indicating groups and periods. Along the top of the periodic table, group numbers are labeled, which are vertical columns of the table. These labels denote the grouping of elements that share similar chemical properties. On the left side, the periods are labeled, which are horizontal rows noted for increasing atomic numbers and changing chemical properties in predictable ways. The layout visually represents the organized structure of elements in the periodic table, showing how groups and periods are organized in a grid-like format. The highlighted labels are meant to aid in understanding the positioning of elements across both dimensions: horizontally as periods and vertically as groups.

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The periodic table is found in section 7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) of the DP chemistry data booklet. Look at the periodic table. Try to locate the element silicon, Si. Notice how it is located in the fourteenth column reading from left to right. This means silicon is in group 14. Notice how silicon is located in the third row reading from top to bottom. This means it is in period 3.

Using the periodic table to help, try to drag and drop these elements, group numbers and period numbers into the correct places.

Interactive 1. Drag and Drop Elements, Group Numbers and Period Numbers Into the Correct Places.

Metals, metalloids and non-metals

Recall from section S2.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/) that we identified the regions of the periodic table where metals and non-metals can be found. Metals are elements on the left-hand side of the periodic table that exhibit the physical properties of malleability, lustre, high melting points, high thermal and electrical conductivities and have a tendency to lose electrons. Non-metals are elements on the right-hand side of the periodic table that are typically brittle, dull and have lower melting points, low thermal and electrical conductivities and have a tendency to gain electrons. All elements can be classified as metals**,** non-metals or metalloids. The majority of elements in the periodic table are metals and are found on the left and the middle of the periodic table.

Metalloids are located in a diagonal staircase that forms a boundary between the metals and non-metals, this is shown in **Figure 3**. Metalloids have properties of both metals and non-metals. They typically have high melting points, intermediate electrical and thermal conductivities, are lustrous and brittle. Silicon for example, is shiny like a metal but a poor conductor of electricity under standard conditions.

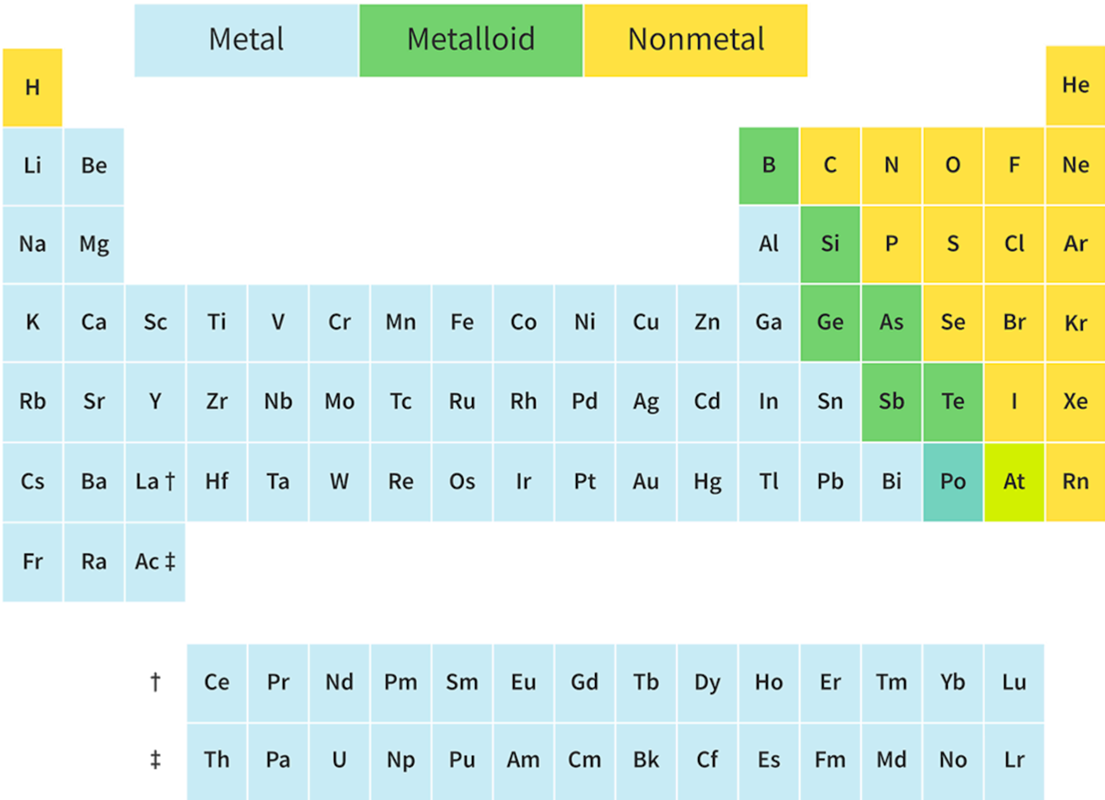

Figure 3. Classification of metals, non-metals and metalloids in the periodic table.

More information for figure 3

The image shows a periodic table with elements classified into metals, nonmetals, and metalloids. The table is color-coded, with metals in blue, nonmetals in yellow, and metalloids in green. Metalloids form a diagonal boundary from boron (B) to astatine (At), positioned between metals and nonmetals. Examples of metalloids include silicon (Si), germanium (Ge), arsenic (As), antimony (Sb), tellurium (Te), and polonium (Po). These elements are often located along a staircase line in the table, illustrating their transitional properties between metals and nonmetals.

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Nature of Science

Aspect: Science as a shared endeavour

There is disagreement in the scientific community about what classifies an element as a metalloid. The elements B, Si, Ge, As, Te, Sb are widely accepted as metalloids but often Bi, Po and At are considered by some scientists as also having intermediate properties between metals and non-metals. Since the pursuit of knowledge in chemistry involves many research scientists this can cause unresolved disagreements during the classification process. These are important to highlight as they bring awareness to the limitations of the classification methods used.

Try to classify each of the elements as a metal, metalloid or non-metal using this drag and drop activity.

Metal	Metalloid	Non-metal

B C Cu I Na Pb Sb Si Xe

Check

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Interactive 2. Classify These Elements as a Metal, Metalloid or Non-Metal.

Blocks in the periodic table

So far, we have considered how to position elements in the periodic table using group number, period number, and the type of element. We can also position elements in the periodic table using the arrangement of their electrons. Can you think about how we might classify elements in the periodic table based on their electron configuration?

Recall from section S1.3.3–4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/main-energy-levels-and-sublevels-id-44043/) that energy levels are divided into s, p, d and f-sublevels. Consider the electron configurations for group 1 elements (H, Li, Na, K, etc.), what sublevel do the outermost valence electrons occupy? What sublevel do the outermost valence electrons occupy for group 2? What about group 17? You might notice that the outermost valence electrons occupy sublevels in a predictable manner. **Table 2** shows the electron configuration for some elements in groups 1, 2 and 17 with the outermost valence electrons emboldened.

Table 1. Electron configurations for groups 1, 2 and

Group 1		Group 2		
H	$\mathbf{1s^1}$			
Li	$1s^2\mathbf{2s^1}$	Be	$1s^2\mathbf{2s^2}$	F
Na	$1s^22s^22p^6\mathbf{3s^1}$	Mg	$1s^22s^22p^6\mathbf{3s^2}$	Cl
K	$1s^22s^22p^63s^23p^6\mathbf{4s^1}$	Ca	$1s^22s^22p^63s^23p^6\mathbf{4s^2}$	Br

This means we can position elements in the periodic table based on the sublevels which the outermost valence electrons occupy. These are called blocks and specify whether an element has the outermost valence electrons occupying the s, p, d or f-sublevels. Hence the regions in the periodic table are referred to as the s-block, p-block, d-block and f-block. These regions are indicated in the periodic table in **Figure 4**. Notice how as you move across period 4 the sublevels the

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outermost valence electrons are occupying move from the 4s to 3d then to 4p. This is due to the 4s-sublevel being slightly lower in energy than the 3d from the convergence of energy levels as you learned about in section S1.3.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/). You will notice the same thing in period 6 moving from 6s to 4f to 5d to 6p.

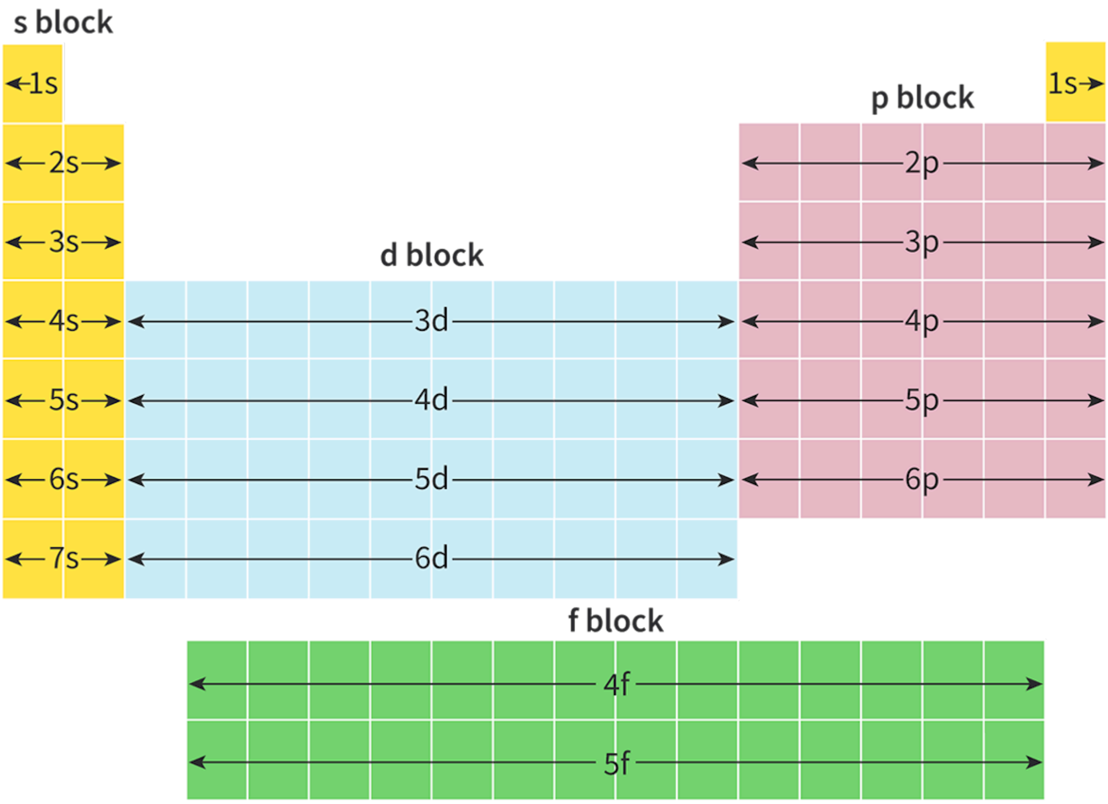

Figure 4. Group and period labels in the periodic table.

More information for figure 4

The image is a diagrammatic representation of the periodic table divided into blocks that indicate where the outermost valence electrons occupy. It shows the layout of the s-block (yellow), p-block (pink), d-block (blue), and f-block (green) across different periods.

The s-block is on the left side, including groups 1 and 2. The p-block is on the right side, spanning groups 13 to 18. The d-block is in the center, covering groups 3 to 12, and consists of transition metals. The f-block is shown as a separate section at the bottom, showcasing the lanthanides and actinides.

Arrows indicate the order in which sublevels are filled within each block, such as 1s, 2s, 3s for the s-block, and 2p, 3p, 4p for the p-block. The d-block arrows display transitions like 3d, 4d, 5d, while the f-block shows 4f and 5f. These directions highlight how electron configurations follow specific patterns when elements are arranged in periods and blocks.

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In **Interactive 3**, use the position in the periodic table to drag and drop the elements into the blocks that indicate where their outermost valence electrons occupy.

s	d	p	f

Ag As Be F Ga Ho K Li

N Ni Pu Sr U W Yb Zr

Check

Interactive 3. Drag and Drop Table Review for Classification of Elements as s, p, d, and f-Block Elements.

Activity

- **IB learner profile attribute:** Knowledgeable

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- **Approaches to learning:** Thinking skills — Asking questions based upon sensible scientific rationale
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair and whole class activity

Element BINGO

You will complete the following activity in groups of 4-6 students.

Part 1: Creation of questions for a BINGO game

- Split up the elements for elements Z = 1 to Z = 54 evenly amongst the group.
- For the elements you are assigned, you will make up a question for each element. Your question could be based on physical properties, chemical properties, structure and bonding, group and period number, electron configuration, or other information. Aim to have a different style of question for each element. For example, what element is a metalloid and has the highest melting point in period 3? If you look at the melting point data in section 8 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/) of the DP chemistry data booklet you would notice that Si has the highest melting point in period 3 and is considered a metalloid.
- Write the question for each element on an individual piece of paper.
- Each group member will fold up their questions and place them in a container. These will be chosen at random later in the BINGO game.

Part 2: Creation of a BINGO card

- Individually, create your own BINGO card.
- Place the symbol for elements of your choosing (from Z = 1 to Z = 54) into each space. You can create your own card or use the template below.

BINGO template (https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Che

Part 3: BINGO game

- As a group you will play BINGO by selecting at random a question inside the container. Each person will take turns selecting the question.
- Individually, try to deduce the element from the clues in the question. Once all members in the group think they know which element it is, share with the group. The person who made up the question can confirm the identity of the element after each group member has shared their guess.
- Once the element has been confirmed, cross the element out on your BINGO (if you have it)
- When you have a full line crossed out (vertical, horizontal, diagonal) call BINGO.

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